MAT120: Integral Calculus

Exercise Set 7.3 — Trigonometric Integrals (Problems 1–22)

Step-by-Step Solutions Grounded in Trigonometric Identities & Reduction Formulas

Formula Reference Guide

Reference Formulas (from Reference Sheets formula1.png, formula2.png, formula3.png)

1. **Integrating Powers of Sine and Cosine (Reduction Formulas)** (`formula1.png`):

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

2. **Integrating Products of Sines and Cosines** $\int \sin^m x\cos^n x\,dx$ (`formula2.png`):

   - **If n is odd:** Split off $\cos x$, use $\cos^2 x = 1-\sin^2 x$, and substitute $u = \sin x$ ($du = \cos x\,dx$).
   - **If m is odd:** Split off $\sin x$, use $\sin^2 x = 1-\cos^2 x$, and substitute $u = \cos x$ ($du = -\sin x\,dx$).
   - **If m and n are both even:** Use half-angle identities:

$$\sin^2 x = \frac{1}{2}(1-\cos 2x), \qquad \cos^2 x = \frac{1}{2}(1+\cos 2x)$$

3. **Product-to-Sum Trigonometric Identities** (`formula3.png`):

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha-\beta)+\sin(\alpha+\beta)]$$

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)]$$

Part I: Indefinite Integrals (Problems 1–16)

Problem 1: Evaluate $\int \cos^2 x \sin x \, dx$

Method / Strategy: `formula2.png` Case 2 ($m = 1$ is odd). Use substitution $u = \cos x$.

Step 1: Let $u = \cos x$. Then $du = -\sin x \, dx$, so $\sin x \, dx = -du$.

Step 2: Substitute into the integral:

$$\int \cos^2 x \sin x \, dx = \int u^2(-du) = -\int u^2 du = -\frac{u^3}{3} + C$$

Step 3: Substitute back $u = \cos x$:

$$\boxed{\int \cos^2 x \sin x \, dx = -\frac{1}{3}\cos^3 x + C}$$

Problem 2: Evaluate $\int \sin^7 3x \cos 3x \, dx$

Method / Strategy: `formula2.png` Case 1 ($n = 1$ is odd). Use substitution $u = \sin 3x$.

Step 1: Let $u = \sin 3x$. Then $du = 3\cos 3x \, dx \implies \cos 3x \, dx = \frac{1}{3}du$.

Step 2: Substitute into the integral:

$$\int \sin^7 3x \cos 3x \, dx = \int u^7 \left(\frac{1}{3}du\right) = \frac{1}{3} \cdot \frac{u^8}{8} + C = \frac{u^8}{24} + C$$

Step 3: Substitute back $u = \sin 3x$:

$$\boxed{\int \sin^7 3x \cos 3x \, dx = \frac{1}{24}\sin^8 3x + C}$$

Problem 3: Evaluate $\int \sin^2 4\theta \, d\theta$

Method / Strategy: `formula2.png` Case 3 (even power). Use half-angle identity $\sin^2 y = \frac{1}{2}(1 - \cos 2y)$ with $y = 4\theta$.

Step 1: Rewrite integrand using double angle:

$$\sin^2 4\theta = \frac{1}{2}(1 - \cos 8\theta)$$

Step 2: Integrate term-by-term:

$$\int \sin^2 4\theta \, d\theta = \frac{1}{2}\int (1 - \cos 8\theta) \, d\theta = \frac{1}{2}\left(\theta - \frac{1}{8}\sin 8\theta\right) + C$$

$$\boxed{\int \sin^2 4\theta \, d\theta = \frac{\theta}{2} - \frac{1}{16}\sin 8\theta + C}$$

Problem 4: Evaluate $\int \cos^2 6x\,dx$

Method / Strategy: `formula2.png` Case 3 (even power). Use half-angle identity $\cos^2 y = \frac{1}{2}(1+\cos 2y)$ with $y = 6x$.

Step 1: Rewrite integrand:

$$\cos^2 6x = \frac{1}{2}(1+\cos 12x)$$

Step 2: Integrate term-by-term:

$$\int \cos^2 6x\,dx = \frac{1}{2}\int (1+\cos 12x)\,dx = \frac{1}{2}\left(x + \frac{1}{12}\sin 12x\right) + C$$

$$\boxed{\int \cos^2 6x\,dx = \frac{x}{2} + \frac{1}{24}\sin 12x + C}$$

Problem 5: Evaluate $\int \sin^3 a\theta\,d\theta \quad (a \neq 0)$

Method / Strategy: `formula2.png` Case 2 ($m = 3$ is odd). Separate one factor of $\sin a\theta$ and use $\sin^2 a\theta = 1 - \cos^2 a\theta$.

Step 1: Rewrite integrand:

$$\sin^3 a\theta = \sin^2 a\theta \cdot \sin a\theta = (1-\cos^2 a\theta)\sin a\theta$$

Step 2: Let $u = \cos a\theta \implies du = -a\sin a\theta\,d\theta \implies \sin a\theta\,d\theta = -\frac{1}{a}du$.

Step 3: Integrate:

$$\int (1-u^2)\left(-\frac{1}{a}du\right) = -\frac{1}{a}\left(u - \frac{u^3}{3}\right) + C = -\frac{1}{a}\cos a\theta + \frac{1}{3a}\cos^3 a\theta + C$$

$$\boxed{\int \sin^3 a\theta\,d\theta = -\frac{1}{a}\cos a\theta + \frac{1}{3a}\cos^3 a\theta + C}$$

Problem 6: Evaluate $\int \cos^3 at\,dt \quad (a \neq 0)$

Method / Strategy: `formula2.png` Case 1 ($n = 3$ is odd). Separate one factor of $\cos at$ and use $\cos^2 at = 1 - \sin^2 at$.

Step 1: Rewrite integrand:

$$\cos^3 at = \cos^2 at \cdot \cos at = (1-\sin^2 at)\cos at$$

Step 2: Let $u = \sin at \implies du = a\cos at\,dt \implies \cos at\,dt = \frac{1}{a}du$.

Step 3: Integrate:

$$\int (1-u^2)\left(\frac{1}{a}du\right) = \frac{1}{a}\left(u - \frac{u^3}{3}\right) + C = \frac{1}{a}\sin at - \frac{1}{3a}\sin^3 at + C$$

$$\boxed{\int \cos^3 at\,dt = \frac{1}{a}\sin at - \frac{1}{3a}\sin^3 at + C}$$

Problem 7: Evaluate $\int \sin bx \cos bx\, dx \quad (b \neq 0)$

Method / Strategy: `formula2.png` Case 1/2 ($m = 1, n = 1$). Use u-substitution $u = \sin bx$.

Step 1: Let $u = \sin bx \implies du = b\cos bx\, dx \implies \cos bx\, dx = \frac{1}{b}du$.

Step 2: Integrate:

$$\int u\left(\frac{1}{b}du\right) = \frac{u^2}{2b} + C = \frac{1}{2b}\sin^2 bx + C$$

$$\boxed{\int \sin bx \cos bx\, dx = \frac{1}{2b}\sin^2 bx + C} \quad \text{(Equivalently, } -\frac{1}{4b}\cos 2bx + C_2\text{)}$$

Problem 8: Evaluate $\int \sin^3 x \cos^3 x\, dx$

Method / Strategy: `formula2.png` Both $m = 3$ and $n = 3$ are odd. Convert $\cos^2 x = 1 - \sin^2 x$ and set $u = \sin x$.

Step 1: Split off one cosine factor:

$$\sin^3 x \cos^3 x = \sin^3 x(1 - \sin^2 x)\cos x$$

Step 2: Substitute $u = \sin x \implies du = \cos x\, dx$:

$$\int u^3(1 - u^2)\, du = \int (u^3 - u^5)\, du = \frac{u^4}{4} - \frac{u^6}{6} + C$$

$$\boxed{\int \sin^3 x \cos^3 x\, dx = \frac{1}{4}\sin^4 x - \frac{1}{6}\sin^6 x + C}$$

Problem 9: Evaluate $\int \sin^2 t \cos^3 t\, dt$

Method / Strategy: `formula2.png` Case 1 ($n = 3$ is odd). Split off $\cos t$ and use $\cos^2 t = 1 - \sin^2 t$.

Step 1: Rewrite integrand:

$$\sin^2 t \cos^3 t = \sin^2 t(1 - \sin^2 t)\cos t$$

Step 2: Substitute $u = \sin t \implies du = \cos t\, dt$:

$$\int u^2(1 - u^2)\, du = \int (u^2 - u^4)\, du = \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$\boxed{\int \sin^2 t \cos^3 t\, dt = \frac{1}{3}\sin^3 t - \frac{1}{5}\sin^5 t + C}$$

Problem 10: Evaluate $\int \sin^3 x \cos^2 x \, dx$

Method / Strategy: `formula2.png` Case 2 ($m = 3$ is odd). Split off $\sin x$ and use $\sin^2 x = 1 - \cos^2 x$.

Step 1: Rewrite integrand:

$$\sin^3 x \cos^2 x = (1 - \cos^2 x) \cos^2 x \sin x$$

Step 2: Substitute $u = \cos x \implies du = -\sin x \, dx$:

$$\int (1 - u^2) u^2 (-du) = \int (u^4 - u^2) \, du = \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$\boxed{\int \sin^3 x \cos^2 x \, dx = \frac{1}{5} \cos^5 x - \frac{1}{3} \cos^3 x + C}$$

Problem 11: Evaluate $\int \sin^2 x \cos^2 x \, dx$

Method / Strategy: `formula2.png` Case 3 ($m = 2, n = 2$ both even). Use half-angle identities.

Step 1: Express in terms of double angles:

$$\sin^2 x \cos^2 x = \left[\frac{1}{2}(1 - \cos 2x)\right]\left[\frac{1}{2}(1 + \cos 2x)\right] = \frac{1}{4}(1 - \cos^2 2x)$$

Step 2: Apply half-angle identity to $\cos^2 2x = \frac{1}{2}(1 + \cos 4x)$:

$$\sin^2 x \cos^2 x = \frac{1}{4}\left[1 - \frac{1}{2}(1 + \cos 4x)\right] = \frac{1}{8}(1 - \cos 4x)$$

Step 3: Integrate:

$$\int \frac{1}{8}(1 - \cos 4x) \, dx = \frac{1}{8}\left(x - \frac{1}{4}\sin 4x\right) + C = \frac{x}{8} - \frac{1}{32}\sin 4x + C$$

$$\boxed{\int \sin^2 x \cos^2 x \, dx = \frac{x}{8} - \frac{1}{32}\sin 4x + C}$$

Problem 12: Evaluate $\int \sin^2 x \cos^4 x \, dx$

Method / Strategy: `formula2.png` Case 3 ($m = 2, n = 4$ both even). Combine half-angle identities.

Step 1: Rewrite integrand:

$$\sin^2 x \cos^4 x = (\sin^2 x \cos^2 x)\cos^2 x = \left[\frac{1}{8}(1 - \cos 4x)\right]\left[\frac{1}{2}(1 + \cos 2x)\right]$$

Alternatively, write as:

$$\sin^2 x \cos^4 x = \frac{1}{8}(1 - \cos 2x)(1 + \cos 2x)^2 = \frac{1}{8}(1 - \cos^2 2x)(1 + \cos 2x) = \frac{1}{8}\sin^2 2x(1 + \cos 2x)$$

Step 2: Split into two manageable integrals:

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{8}\int \sin^2 2x \, dx + \frac{1}{8}\int \sin^2 2x \cos 2x \, dx$$

Step 3: Evaluate each component:

- $\frac{1}{8}\int \frac{1}{2}(1 - \cos 4x)\, dx = \frac{1}{16}\left(x - \frac{1}{4}\sin 4x\right) = \frac{x}{16} - \frac{1}{64}\sin 4x$
- $\frac{1}{8}\int \sin^2 2x \cos 2x \, dx$: let $u = \sin 2x \implies du = 2\cos 2x \, dx \implies \frac{1}{16}\int u^2 du = \frac{1}{48}\sin^3 2x$

$$\boxed{\int \sin^2 x \cos^4 x \, dx = \frac{x}{16} - \frac{1}{64}\sin 4x + \frac{1}{48}\sin^3 2x + C}$$

Problem 13: Evaluate $\int \sin 3x \cos 4x \, dx$

Method / Strategy: `formula3.png` Use Product-to-Sum formula: $2\sin\alpha\cos\beta = \sin(\alpha - \beta) + \sin(\alpha + \beta)$.

Step 1: Set $\alpha = 3x, \beta = 4x$:

$$\sin 3x \cos 4x = \frac{1}{2}[\sin(-x) + \sin 7x] = \frac{1}{2}[\sin 7x - \sin x]$$

Step 2: Integrate term-by-term:

$$\int \frac{1}{2}(\sin 7x - \sin x)\, dx = \frac{1}{2}\left(-\frac{1}{7}\cos 7x + \cos x\right) + C$$

$$\boxed{\int \sin 3x \cos 4x \, dx = \frac{1}{2}\cos x - \frac{1}{14}\cos 7x + C}$$

Problem 14: Evaluate $\int \sin 3\theta \cos 2\theta \, d\theta$

Method / Strategy: `formula3.png` Use Product-to-Sum formula with $\alpha = 3\theta, \beta = 2\theta$.

Step 1: Apply Product-to-Sum identity:

$$\sin 3\theta \cos 2\theta = \frac{1}{2}[\sin(\theta) + \sin(5\theta)]$$

Step 2: Integrate:

$$\int \frac{1}{2}(\sin\theta + \sin 5\theta)\, d\theta = \frac{1}{2}\left(-\cos\theta - \frac{1}{5}\cos 5\theta\right) + C$$

$$\boxed{\int \sin 3\theta \cos 2\theta \, d\theta = -\frac{1}{2}\cos\theta - \frac{1}{10}\cos 5\theta + C}$$

Problem 15: Evaluate $\int \sin x \cos\left(\frac{x}{2}\right) dx$

Method / Strategy: `formula3.png` Product-to-Sum with $\alpha = x, \beta = x/2$.

Step 1: Apply identity:

$$\sin x \cos\left(\frac{x}{2}\right) = \frac{1}{2}\left[\sin\left(\frac{x}{2}\right) + \sin\left(\frac{3x}{2}\right)\right]$$

Step 2: Integrate:

$$\int \frac{1}{2}\left[\sin\left(\frac{x}{2}\right) + \sin\left(\frac{3x}{2}\right)\right] dx = \frac{1}{2}\left[-2\cos\left(\frac{x}{2}\right) - \frac{2}{3}\cos\left(\frac{3x}{2}\right)\right] + C$$

$$\boxed{\int \sin x \cos\left(\frac{x}{2}\right) dx = -\cos\left(\frac{x}{2}\right) - \frac{1}{3}\cos\left(\frac{3x}{2}\right) + C}$$

Problem 16: Evaluate $\int \cos^{1/3} x \sin x \, dx$

Method / Strategy: Direct substitution $u = \cos x$.

Step 1: Let $u = \cos x \implies du = -\sin x \, dx \implies \sin x \, dx = -du$.

Step 2: Integrate power function:

$$\int u^{1/3}(-du) = -\frac{u^{4/3}}{4/3} + C = -\frac{3}{4}u^{4/3} + C$$

$$\boxed{\int \cos^{1/3} x \sin x \, dx = -\frac{3}{4}\cos^{4/3} x + C}$$

Part II: Definite Integrals (Problems 17–22)

Problem 17: Evaluate $\int_0^{\pi/2} \cos^3 x\,dx$

Method / Strategy: `formula2.png` Case 1 ($n = 3$ odd). Use $\cos^2 x = 1 - \sin^2 x$ and $u = \sin x$.

Step 1: Rewrite integrand:

$$\int_0^{\pi/2} (1 - \sin^2 x)\cos x\,dx$$

Step 2: Substitute $u = \sin x \implies du = \cos x\,dx$. Update limits:

- $x = 0 \implies u = \sin 0 = 0$
- $x = \pi/2 \implies u = \sin(\pi/2) = 1$

Step 3: Evaluate:

$$\int_0^1 (1 - u^2)\,du = \left[u - \frac{u^3}{3}\right]_0^1 = \left(1 - \frac{1}{3}\right) - (0) = \frac{2}{3}$$

$$\boxed{\int_0^{\pi/2} \cos^3 x\,dx = \frac{2}{3}}$$

Problem 18: Evaluate $\int_0^{\pi/2} \sin^2\frac{x}{2}\cos^2\frac{x}{2}\,dx$

Method / Strategy: Simplify using double angle $\sin\frac{x}{2}\cos\frac{x}{2} = \frac{1}{2}\sin x$, then apply `formula2.png` half-angle identity.

Step 1: Simplify integrand:

$$\sin^2\frac{x}{2}\cos^2\frac{x}{2} = \left(\frac{1}{2}\sin x\right)^2 = \frac{1}{4}\sin^2 x = \frac{1}{8}(1 - \cos 2x)$$

Step 2: Integrate and evaluate limits:

$$\frac{1}{8}\int_0^{\pi/2} (1 - \cos 2x)\,dx = \frac{1}{8}\left[x - \frac{1}{2}\sin 2x\right]_0^{\pi/2}$$

Step 3: Substitute limits:

$$= \frac{1}{8}\left[\left(\frac{\pi}{2} - \frac{1}{2}\sin\pi\right) - (0 - 0)\right] = \frac{1}{8}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{16}$$

$$\boxed{\int_0^{\pi/2} \sin^2\frac{x}{2}\cos^2\frac{x}{2}\,dx = \frac{\pi}{16}}$$

Problem 19: Evaluate $\int_0^{\pi/3} \sin^6 3x \cos^3 3x \, dx$

Method / Strategy: `formula2.png` Case 1 ($n = 3$ odd). Substitute $u = \sin 3x$.

Step 1: Substitute $u = \sin 3x \implies du = 3\cos 3x \, dx$.

Step 2: Change limits of integration:

- $x = 0 \implies u = \sin(0) = 0$
- $x = \pi/3 \implies u = \sin(3 \cdot \pi/3) = \sin(\pi) = 0$

Step 3: Since lower and upper limits are identical ($u = 0$ to $u = 0$):

$$\int_0^0 \ldots du = 0$$

$$\boxed{\int_0^{\pi/3} \sin^6 3x \cos^3 3x \, dx = 0}$$

Problem 20: Evaluate $\int_{-\pi}^{\pi} \cos^2 4\theta \, d\theta$

Method / Strategy: `formula2.png` Half-angle identity $\cos^2 4\theta = \frac{1}{2}(1 + \cos 8\theta)$.

Step 1: Rewrite integral:

$$\int_{-\pi}^{\pi} \frac{1}{2}(1 + \cos 8\theta) \, d\theta = \frac{1}{2}\left[\theta + \frac{1}{8}\sin 8\theta\right]_{-\pi}^{\pi}$$

Step 2: Evaluate limits:

$$= \frac{1}{2}\left[\left(\pi + \frac{1}{8}\sin 8\pi\right) - \left(-\pi + \frac{1}{8}\sin(-8\pi)\right)\right] = \frac{1}{2}[\pi - (-\pi)] = \pi$$

$$\boxed{\int_{-\pi}^{\pi} \cos^2 4\theta \, d\theta = \pi}$$

Problem 21: Evaluate $\int_0^{\pi/3} \sin 4x \cos 2x \, dx$

Method / Strategy: `formula3.png` Product-to-Sum identity $\sin 4x \cos 2x = \frac{1}{2}[\sin 2x + \sin 6x]$.

Step 1: Rewrite integral:

$$\int_0^{\pi/3} \frac{1}{2}(\sin 2x + \sin 6x)\, dx = \frac{1}{2}\left[-\frac{1}{2}\cos 2x - \frac{1}{6}\cos 6x\right]_0^{\pi/3}$$

Step 2: Evaluate at upper limit $x = \pi/3$:

$$-\frac{1}{2}\cos\left(\frac{2\pi}{3}\right) - \frac{1}{6}\cos(2\pi) = -\frac{1}{2}\left(-\frac{1}{2}\right) - \frac{1}{6}(1) = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

Step 3: Evaluate at lower limit $x = 0$:

$$-\frac{1}{2}\cos(0) - \frac{1}{6}\cos(0) = -\frac{1}{2} - \frac{1}{6} = -\frac{2}{3}$$

Step 4: Subtract lower from upper and multiply by $1/2$:

$$= \frac{1}{2}\left[\frac{1}{12} - \left(-\frac{2}{3}\right)\right] = \frac{1}{2}\left[\frac{1}{12} + \frac{8}{12}\right] = \frac{1}{2} \cdot \frac{9}{12} = \frac{3}{8}$$

$$\boxed{\int_0^{\pi/3} \sin 4x \cos 2x \, dx = \frac{3}{8}}$$

Problem 22: Evaluate $\int_0^{2\pi} \sin^2 kx \, dx \quad (k \in \mathbb{Z} \setminus \{0\})$

Method / Strategy: `formula2.png` Half-angle identity $\sin^2 kx = \frac{1}{2}(1 - \cos 2kx)$.

Step 1: Rewrite integrand:

$$\int_0^{2\pi} \frac{1}{2}(1 - \cos 2kx)\, dx = \frac{1}{2}\left[x - \frac{1}{2k}\sin 2kx\right]_0^{2\pi}$$

Step 2: Evaluate for non-zero integer k:

$$= \frac{1}{2}\left[\left(2\pi - \frac{1}{2k}\sin(4\pi k)\right) - (0 - 0)\right]$$

Since $\sin(4\pi k) = 0$ for any integer k:

$$= \frac{1}{2}(2\pi) = \pi$$

$$\boxed{\int_0^{2\pi} \sin^2 kx \, dx = \pi \quad \text{for non-zero integer } k}$$